

PolyAgora Meta-Regime Steering Layer

CTO Technical Specification V0.1

PolyAgora / PolygonEye

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1 Strategic Objective

The next architectural phase of PolyAgora is not to replace existing strategies (e.g. Momentum), but to introduce a higher-order runtime steering layer capable of dynamically allocating among multiple admissible strategy manifolds.

The objective shifts from:

- relative benchmark outperformance

toward:

- attractive absolute returns
- recoverability
- maneuverability
- institutional-grade robustness
- low correlation to traditional asset classes

in line with institutional requirements such as:

- Sharpe > 1.5
- Sortino > 1.5
- Calmar > 1.5
- low equity/bond correlation
- robust live-trading survivability

2 Core Concept

We construct a two-level architecture.

2.1 Level 1: Strategy Manifolds

Existing strategy manifolds:

- PolyAgora V7
- Momentum
- Defensive / CHASH

Each remains independently defined and frozen initially.

2.2 Level 2: Meta Steering Layer

A higher-order MRTP-driven steering layer dynamically allocates among those manifolds.

This is not:

“Momentum versus PolyAgora”

but:

“PolyAgora governing admissibility of strategy geometries.”

3 Conceptual Interpretation

Each strategy becomes:

- a regime-space
- a maneuverability geometry
- a recoverability profile

3.1 Momentum

Strong under:

- persistent trends
- coherent low-volatility expansions

Weak under:

- violent reversals
- fragmentation
- transition shocks

3.2 PolyAgora V7

Strong under:

- transitions
- fragmentation
- boundary instability
- recovery phases
- regime deformation

3.3 Defensive / CHASH

Strong under:

- rupture
- inadmissibility
- collapse geometry

4 Layer Separation Principle

The meta-layer must remain:

- slower
- coarser
- lower entropy

- governance-oriented

than underlying manifolds.

The architecture must avoid:

- recursive steering instability
- oscillatory reflexivity
- self-referential feedback loops
- hyper-reactive switching

5 Minimal Architecture

5.1 Asset Layer

Existing:

- assets
- signals
- returns
- weights

No change.

5.2 Strategy Manifold Layer

Each manifold outputs:

- asset weights
- expected returns
- volatility profile
- drawdown profile
- turnover
- admissibility diagnostics

Examples:

$$M_1 = \text{PolyAgora V7}$$

$$M_2 = \text{Momentum}$$

$$M_3 = \text{Defensive}$$

5.3 Meta Steering Layer

Inputs:

- regime persistence
- volatility stress
- boundary tension
- recoverability
- admissibility confidence
- transition density

Outputs:

- manifold allocations

6 Meta State Definition

Define:

$$\mathcal{R}_t = (r_t, v_t, c_t, b_t)$$

Where:

- r_t = regime persistence
- v_t = volatility stress
- c_t = admissibility confidence
- b_t = boundary tension

Potential sources include:

- VIX
- survival matrix entropy
- zone transitions
- drawdown acceleration
- realized volatility
- VAIDM / ADD overlays

7 Strategy Admissibility

Each manifold receives:

$$A_i(t) \in [0, 1]$$

representing:

“How admissible is this strategy manifold under current geometry?”

8 MRTP Meta Steering

For each manifold:

$$MRTP_i(t) = \alpha S_i(t) + \beta F_i(t) - \gamma D_i(t)$$

Where:

- $S_i(t)$ = persistence stability
- $F_i(t)$ = future maneuverability
- $D_i(t)$ = expected damage if wrong

Interpretation:

MRTP does not maximize immediate return.

MRTP maximizes:

- future reachable admissible states
- recoverability
- maneuverability preservation

9 Meta Allocation

Define manifold allocation:

$$W_i(t) = \frac{A_i(t) \exp(\lambda MRTP_i(t))}{\sum_j A_j(t) \exp(\lambda MRTP_j(t))}$$

Where:

- $W_i(t)$ = allocation to manifold i
- λ = aggressiveness parameter

This produces:

- smooth transitions
- probabilistic steering
- no hard switching

10 Final Portfolio

Each manifold produces:

$$w_i^{assets}(t)$$

Final portfolio:

$$w^{final}(t) = \sum_i W_i(t) \cdot w_i^{assets}(t)$$

11 Runtime Timing

Recompute:

- daily
- 16:00 ET close

Using:

- forward-safe information only
- no same-bar contamination
- no hindsight

12 Validation Protocol

Critical: The meta-layer must not see future returns.

At time t , it only sees:

$$\mathcal{F}_{t-1}$$

Meaning:

- realized information only
- no future manifold performance

12.1 Freeze Manifolds

Initial versions remain fixed:

- V7
- Momentum
- Defensive

No online mutation initially.

12.2 Walk-Forward Validation

Example:

- Train: 2015–2019
- Test: 2020

Then roll forward.

No global optimization.

12.3 Crisis Validation

Test specifically:

- COVID
- inflation transitions
- volatility spikes
- trend crashes
- reversal regimes

This architecture is fundamentally about transition maneuverability.

12.4 Meta Advantage Validation

Compare:

1. Static Momentum
2. Static PolyAgora
3. Meta-Steered System

Measure:

- Sharpe
- Sortino
- Calmar
- drawdown
- recoverability
- turnover
- switching frequency
- correlation profile

13 Meta-Level Fixed Star / Survival Geometry

The meta-layer itself requires:

- survival structure
- admissibility topology
- higher-order Fixed Stars

The objects are not assets.

The objects are:

- strategy manifolds

13.1 Meta Survival Matrix

Define:

$$S_{ij}^{meta}$$

Meaning:

“Admissibility / survival relation between strategy manifolds.”

Examples:

- Momentum survives persistent expansions
- PolyAgora survives fragmentation
- Defensive survives collapse

This may create:

- local stars
- manifold blocks
- strategic geometries

14 Important Distinction

Meta Fixed Star is not:

- best-performing strategy

It is:

- stable maneuverability center under current geometry

This distinction is essential to avoid:

- hindsight optimization
- performance chasing
- hidden overfit

15 Reference Polygon (RP)

15.1 Initial Recommendation

Use the same RP for:

- asset level
- meta level

Reason:

- coherence
- semantic alignment
- stable admissibility structure

15.2 Long-Term Possibility

Eventually: the meta layer may evolve toward a more abstract RP describing:

- persistence
- recoverability
- convexity
- fragmentation
- maneuverability

while the asset-layer RP continues describing:

- market geometry
- sector structure
- macro relations

16 Critical Constraints

Do not:

- recursively steer steering layers
- add meta-meta learning
- add adaptive manifold mutation
- create high-frequency switching
- create hyper-reactive transitions

17 Initial Implementation Recommendation

Initial version should only use:

- PolyAgora V7
- Momentum
- Defensive / CHASH

Do not yet add:

- carry
- convexity overlays
- macro engines
- online learning

First validate:

- manifold steering stability
- recoverability advantage
- admissibility quality
- transition handling

18 Strategic Interpretation

PolyAgora is evolving from:

- a strategy engine

toward:

- a runtime governance architecture over strategy spaces

The system no longer merely optimizes positions.

It governs:

- admissibility
- maneuverability
- recoverability
- strategy transitions

under changing market geometries.